PROJECT REPORT

BHARAT VR: IMMERSIVE 360° TOURISM

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🔗 GitHub: https://github.com/CodeWizardAn/BharatVR

1. Introduction

BharatVR is a full-stack web application designed to allow users to virtually explore iconic Indian tourist destinations through immersive 360° Street View experiences. Directly accessible from any modern web browser, the platform eliminates the need for specialized VR hardware, making high-end virtual tourism accessible to the masses.

1.1 Project Objective

The primary goal of BharatVR is to promote Indian heritage by providing a cinematic, interactive experience of landmarks, beaches, and natural wonders. It serves as a comprehensive travel guide, offering logistical details and community feedback alongside the visual experience.

2. System Interface & Screenshots

Figure 1: Homepage - Dark luxury design with parallax Taj Mahal backdrop.

Figure 2: Destination Browser - Grid view with search and category filters.

Figure 3: VR Viewer - Full 360° panoramic immersion for the user.

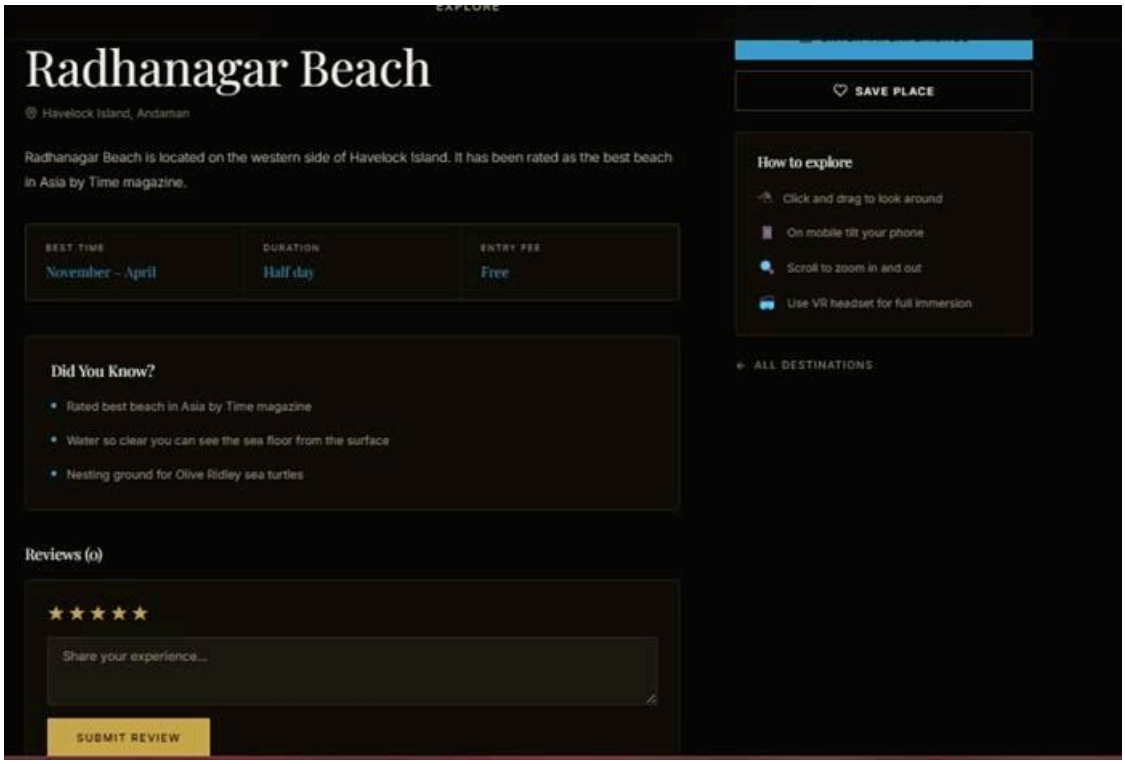

Figure 4: Destination Detail - Facts, entry fees, and review system.

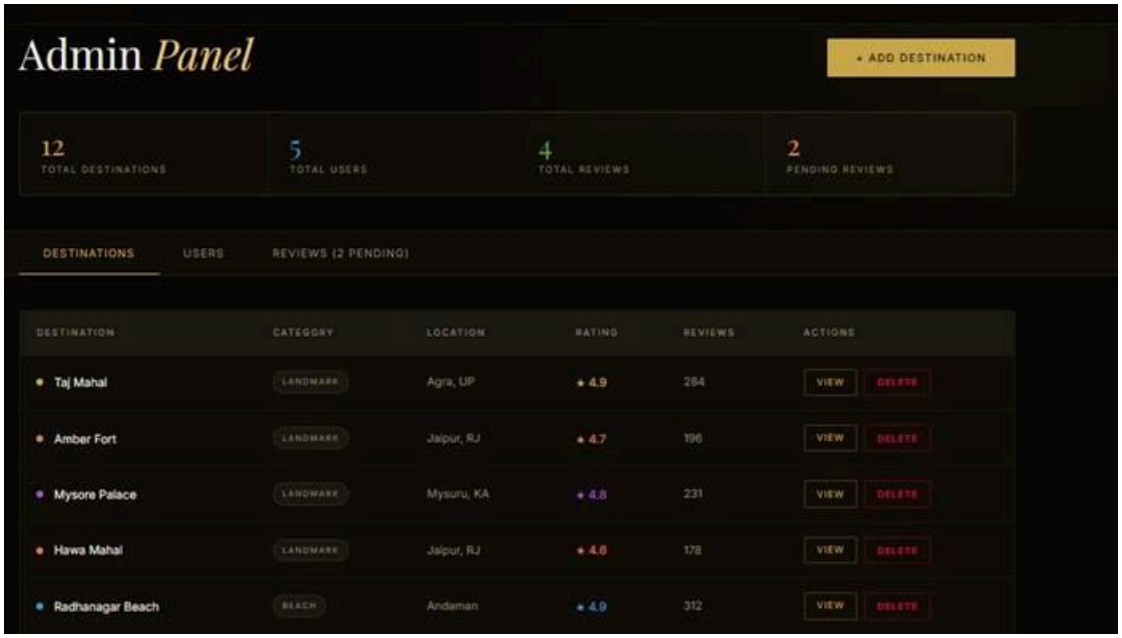

Figure 5: Admin Panel - Dashboard for content management and review moderation.

3. Features & Functionalities

3.1 Core Functionality

- 360° VR Experience: Google Street View panoramas for all 12 destinations.
- Destination Browser: Real-time search and filtering (Landmark, Beach, Mountain).
- Authentication: JWT-based secure login/register with password hashing.
- User Profile: Management of saved 'favourites' and personal review history.
- Admin Control: Full CRUD for destinations and moderation tools for reviews.

3.2 UI/UX Enhancements

- Cinematic Design: Dark luxury theme with gold accents and elegant typography.
- Smooth Motion: Framer Motion transitions and parallax scrolling.
- Skeleton Loading: Graceful handling of asynchronous data fetching.

- Responsive Design: Fully functional across desktop, tablet, and mobile browsers.

4. Technology Stack

Component	Technologies Used
Frontend	React, Vite, React Router, Framer Motion, Axios, A-Frame.js
Backend	Node.js, Express.js, JWT, bcryptjs, Multer
Database	MongoDB Atlas, Mongoose ODM

5. Project Structure & Architecture

The project follows a standard MERN architecture with a separated frontend and backend.

```
bharatvr/
 ├── frontend/ (React + Vite)
 │   ├── src/api/ (Axios calls)
 │   ├── src/components/ (Navbar, Footer)
 │   ├── src/context/ (Auth State)
 │   └── src/pages/ (Home, VR Viewer, Admin)
 └── backend/ (Node + Express)
        ├── models/ (User, Destination, Review)
        ├── routes/ (Auth, Reviews, Users)
        └── middleware/ (Auth protection)
```

6. Destinations Covered

Category	Location	State
Landmark	Taj Mahal	Uttar Pradesh
Landmark	Amber Fort	Rajasthan
Beach	Radhanagar	Andaman
Beach	Palolem	Goa

Nature	Munnar	Kerala
Nature	Rohtang Pass	Himachal

7. Conclusion & Future Scope

BharatVR successfully demonstrates the integration of modern web technologies with VR standards to create a compelling tourism product. Future enhancements include live weather integration, YouTube overview videos, and PWA support for offline access.